Residue Theorem and Applications

Complex Analysis

Review: Isolated Singularities

Let f be analytic except at isolated points.
A point z_0 is an **isolated singularity** if

$$0 < |z - z_0| < r$$

is a region where f is analytic.
Laurent expansion:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

The coefficient

$$a_{-1}$$

is called the **residue** of f at z_0.

$$\mathrm{Res}(f, z_0) = a_{-1}$$

Residue Theorem

Let f be analytic inside and on a simple closed contour C except at finitely many isolated singularities $z_1, \ldots, z_n$ inside C.
Then

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \mathrm{Res}(f, z_k)$$

Thus a complicated contour integral reduces to computing the residues of the enclosed singularities.

Residue at a Simple Pole

If z_0 is a **simple pole**,

$$f(z) = \frac{\phi(z)}{z - z_0}$$

where $\phi(z)$ is analytic and $\phi(z_0) \neq 0$.
Then

$$\text{Res}(f, z_0) = \lim_{z \to z_0} (z - z_0) f(z)$$

Residue at a Pole of Order m

If z_0 is a pole of order m,

$$\mathrm{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z - z_0)^m f(z) \right]$$

In practice we most often encounter

- simple poles
- rational functions

which make residue computations straightforward.

Residues of Rational Functions

Suppose

$$f(z) = \frac{p(z)}{q(z)}$$

and z_0 is a simple zero of q. Then

$$\operatorname{Res}(f, z_0) = \frac{p(z_0)}{q'(z_0)}$$

This formula is extremely useful for integrals involving rational functions.

Improper Integrals on $(-\infty, \infty)$

Suppose

$$I = \int_{-\infty}^{\infty} R(x)\,dx$$

where R is a rational function.
Method:

1. Extend $R(x)$ to $R(z)$.
2. Integrate over a large semicircle in the upper half-plane.
3. Use the residue theorem.
4. Let the radius $\to \infty$.

If the semicircle contribution vanishes,

$$\int_{-\infty}^{\infty} R(x)\,dx = 2\pi i \sum \operatorname{Res}(R, z_k)$$

where z_k are poles in the upper half-plane.

Example

Evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$$

Poles:

$$z = \pm i$$

Upper half-plane pole:

$$z = i$$

Residue:

$$\operatorname{Res}\left(\frac{1}{z^2+1}, i\right) = \lim_{z \to i} \frac{z-i}{(z-i)(z+i)} = \frac{1}{2i}$$

Residue theorem gives

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = 2\pi i \left(\frac{1}{2i}\right) = \pi$$

Example

Evaluate

$$\int_{-\infty}^{\infty} \frac{dx}{x^4+1}$$

Factor denominator:

$$x^4+1=(x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$$

Poles in upper half-plane:

$$z=\frac{1+i}{\sqrt{2}}, \quad z=\frac{-1+i}{\sqrt{2}}$$

Compute residues and apply

$$\int_{-\infty}^{\infty} \frac{dx}{x^4+1}=\frac{\pi}{\sqrt{2}}$$

Trigonometric Integrals

Residues also evaluate integrals of the form

$$\int_0^{2\pi} F(\cos\theta, \sin\theta)\, d\theta$$

Use the substitution

$$z = e^{i\theta}$$

Then

$$d\theta = \frac{dz}{iz}$$

and

$$\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

$$\sin\theta = \frac{1}{2i}\left(z - \frac{1}{z}\right)$$

Resulting Contour Integral

The integral becomes

$$\int_0^{2\pi} F(\cos\theta, \sin\theta)\, d\theta = \oint_{|z|=1} F\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right) \frac{dz}{iz}$$

Thus the integral is transformed into a contour integral on the **unit circle**.
Residues inside $|z|<1$ determine the value.

Example

Evaluate

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$$

Using

$$\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

the integral becomes a rational function in z integrated around $|z| = 1$.
Residue calculation yields

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \frac{2\pi}{\sqrt{3}}$$

Summary

Residue theorem:

$$\oint_C f(z)\,dz = 2\pi i \sum \mathrm{Res}(f, z_k)$$

Residues can be computed using:

- Laurent coefficient a_{-1}
- limit formula for simple poles
- derivative formula for higher poles
- $\dfrac{p(z_0)}{q'(z_0)}$ for rational functions with simple poles.

Applications:

- improper integrals on $(-\infty, \infty)$
- trigonometric integrals via $z = e^{i\theta}$